GEOptimize

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Proposal for:	Proposal Date:	Proposal Number:
Matt Ferris Project Manager 555 Penbrooke Dr Penfield, NY 14526	2026-04-06	QU-0225 **Reference** US NY Popli Blake Hall

Understanding of Project

Popli Design Group is undertaking the rehabilitation of the A Wing of Blake Hall at SUNY Geneseo, located in Geneseo, New York. The project involves the renovation of an approximately 27,195 square foot, four-story building that will continue to serve as an administrative office and multicultural center. The facility includes a mix of office spaces, conference rooms, and light recreational areas, with no dormitory, laboratory, or large classroom functions.

As part of the renovation, the building envelope is being upgraded with new wall insulation and windows, while the roof was previously replaced in 2022. In parallel, Popli Design Group is designing a new all-electric mechanical system, including heating and cooling systems supported by heat recovery ventilation, as well as updated plumbing systems.

The project team is evaluating the potential implementation of a ground source heat pump (GSHP) system to support the building's heating and cooling needs. Several nearby open field areas, located approximately 1,000 feet from the building, have been identified as potential locations for a geothermal well field; however, site suitability has not yet been confirmed, and no well drilling logs are currently available.

Popli Design Group has engaged GEOptimize to provide geothermal design support services for this project. This effort will include evaluating site feasibility and supporting ground heat exchanger (GHX) design.

Scope of Work

Preliminary GHX Sizing

GEOptimize will develop a preliminary GHX design using a prototypical building energy model and Ground Loop Design (GLD) software to estimate heating and cooling loads. Sizing will be informed by available geotechnical data, regional geology, and coordination with local drilling contractors to reflect site-specific conditions.

Relevant design and performance data from the Doty Hall geothermal system will also be reviewed, if available, to help calibrate assumptions. The result will be a conceptual GHX layout and sizing basis to support early-stage design decisions and identify key parameters for further refinement.

Energy Model Review

Popli Design Group will be responsible for developing the building energy model for the project. GEOptimize will review the resulting 8,760-hour heating and cooling load profiles, along with key modeling inputs and assumptions, including building envelope characteristics, ventilation requirements, and system configurations.

This review will include a comprehensive quality assurance and benchmarking assessment, informed by industry experience and comparable projects, to confirm that the model outputs are appropriate for use in GHX sizing and overall geothermal system design.

GSHP System Design Review

GEOptimize will review the GSHP system design, including Ground Loop Design (GLD) or LoopLinkPro modeling inputs and outputs. This review will encompass key parameters such as subsurface thermal properties, borefield configuration, predicted source temperatures, and mechanical system selections.

The objective of this review is to validate design assumptions, ensure consistency with project requirements, and confirm that the proposed system configuration is appropriate for reliable and efficient operation.

Construction Drawing Package Review

GEOptimize will review and provide revisions to the construction drawing package and technical specifications for the geothermal system. This review will include key system components and design elements, including circulation pumps, heat pump selection, GHX field header configuration, hydronic schematics, and a sequence of operation.

The objective of this effort is to ensure that the design is technically sound, well-coordinated, and aligned with best practices for performance, constructability, and long-term system reliability.

Construction Administration Support

GEOptimize will assist with the following items during the construction administration phase:

- Interview contractors for construction of the GHX
- Conduct a pre-construction virtual meeting with the selected mechanical contractor, geothermal heat pump contractor, GHX contractor, and project manager to coordinate on-site construction and excavation
- Review shop drawings to verify materials and equipment meet contract requirements
- Review and take appropriate action on contractor submittals
- Respond to contractor Requests for Information (RFIs)
- Prepare and issue supplemental instructions to clarify contract documents as needed
- Prepare proposed change notices (including drawings and specifications), evaluate contractor proposals, and issue change directives and change orders
- Review close-out submittals
- Review operation and maintenance manuals
- Provide warranty support

Information Required

- Energy model inputs and outputs
- Site plan indicating land area available for construction of the GHX.
- Geotechnical report.
- GSHP system construction drawings.

Deliverables

- Preliminary GHX sizing analysis in PowerPoint format.
- Coordination emails and virtual meetings.

Pricing

Description			Line Total
Preliminary GHX Sizing			$4,450.00
Energy Model Review			$7,450.00
GSHP System Design Review			$7,950.00
Construction Drawing Package Review			$4,950.00
Construction Administration Support			$5,150.00
Total (USD)			**$29,950.00**

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